E-Commerce Data Visualization Report

18 Standard Plots & Analysis

Generated: April 08, 2026

Dataset: 15,050 transactions | 14 columns

```
This report contains:
• 18 Standard Visualization Types
• Data Cleaning & Handling Missing Values
• Statistical Insights
• Code Samples for Each Plot

Categories Covered:
✓ Distribution Analysis (Histograms, KDE, Box Plots)
✓ Group Comparisons (Violin Plots, Grouped Boxes)
✓ Relationships (Scatter Plots, Correlation Heatmaps)
✓ Categorical Analysis (Count Plots, Bar Charts, Pie Charts)
✓ Multi-Dimensional Views (PairPlots, Dashboards)
```

TABLE OF CONTENTS

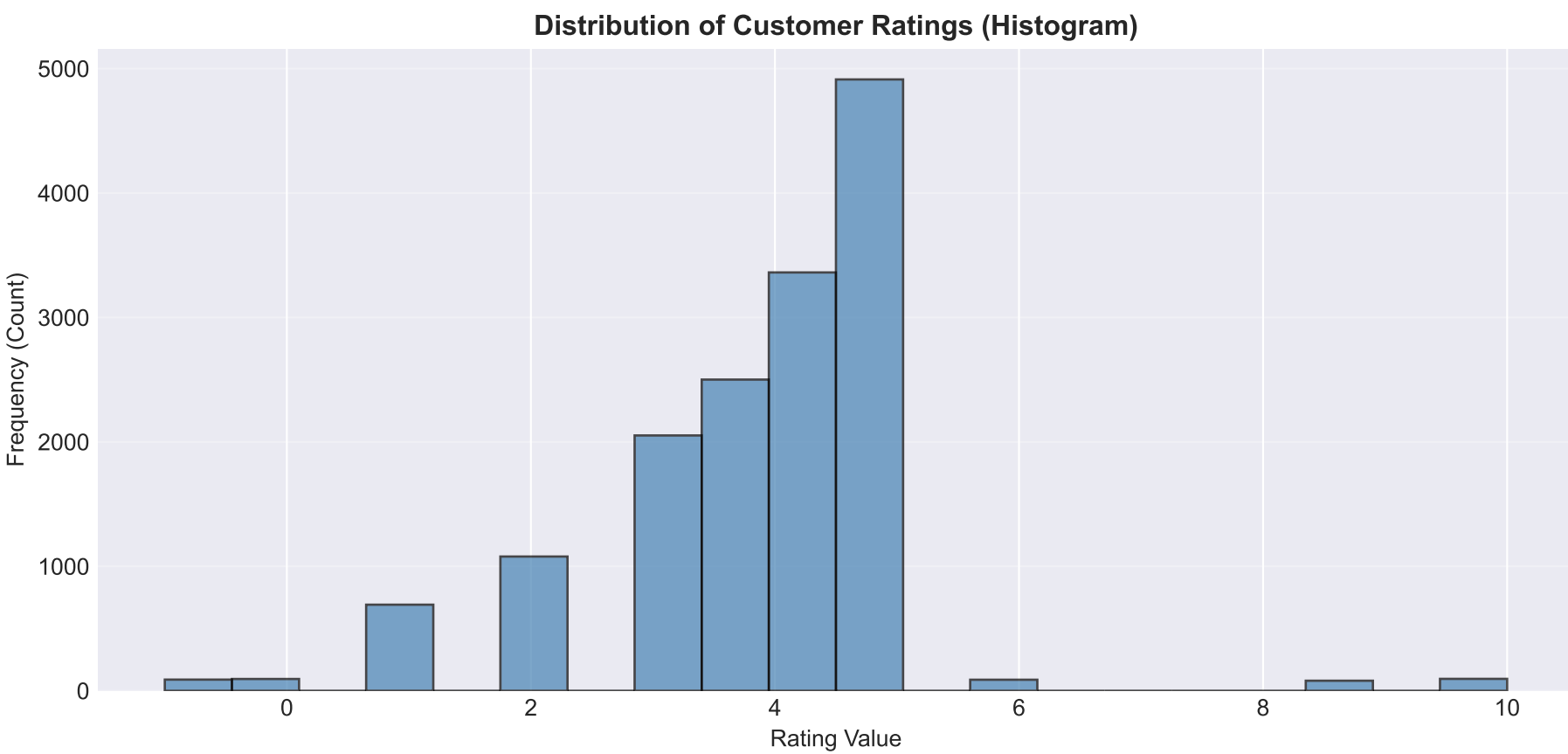
1. Histogram - Distribution of Ratings
2. Histogram with KDE - Ratings with Smooth Curve
3. Box Plot - Rating Spread & Outliers
4. Grouped Box Plot - Ratings by Delivery Status
5. Violin Plot - Detailed Distribution by Group
6. Count Plot - Order Frequency by Status
7. Bar Chart - Average Rating by Status
8. Grouped Bar Chart - Payment × Delivery Status
9. Scatter Plot - Quantity vs Rating
10. Scatter with Hue - Colored by Delivery Status
11. Scatter with Colorbar - Price vs Amount (by Quantity)
12. Line Plot - Average Rating by Category
13. Pie Chart - Payment Method Distribution
14. Donut Chart - Delivery Status Distribution
15. Correlation Heatmap - Numeric Columns Relationships
16. PairPlot - All Numeric Columns Grid
17. Comparison - Missing Values Impact
18. Dashboard - 9-Panel Comprehensive View

Plus: Data Cleaning Summary & Statistical Insights

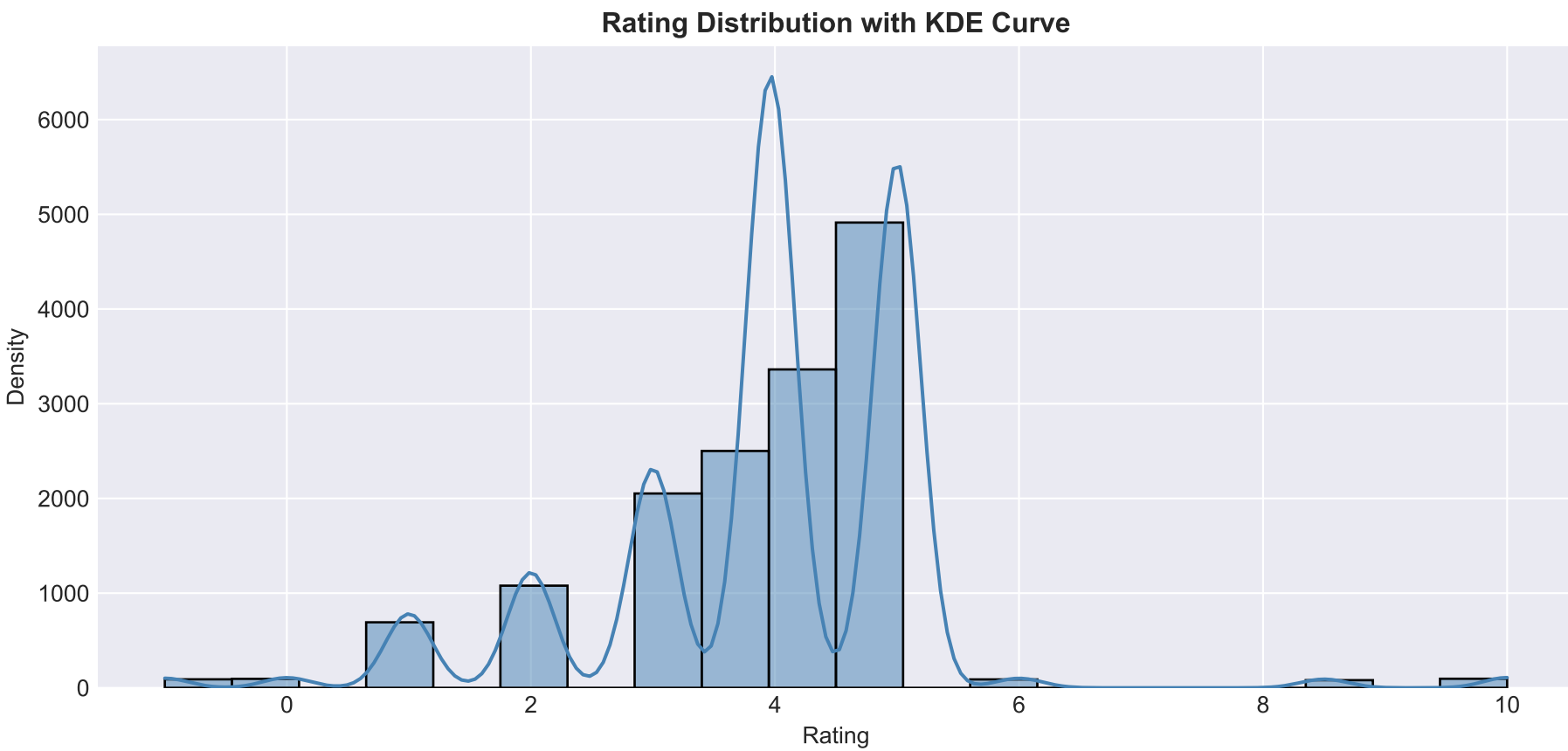
DATA CLEANING SUMMARY

```
    ORIGINAL DATASET:
    • Total Records: 15,050
    • Total Columns: 14

    MISSING VALUES HANDLED:
    • Customer ID: Filled with 'Unknown'
    • City: Filled with Mode (Most Frequent)
    • Product Name: Cleaned (lowercase, stripped) & filled with Mode
    • Rating: Filled with Mean (3.91)
    • Delivery Status: Filled based on Payment Method logic

    DATA TRANSFORMATIONS:
    • Rating Categorization: Excellent (5+), Very Good (4+), Good (3+), Average (2+), Bad (1+)
    • Delivery Status Logic:
      - If CASH payment → 'Shipped'
      - If Card payment → 'Pending'
      - Otherwise → 'Processing'

    CURRENT DATA QUALITY:
    • Total Non-Null Records: 15,050
    • Duplicate Rows: 14
    • Data Types: {<StringDtype(storage='python', na_value=nan)>: 9, dtype('float64'): 5}

    NUMERIC COLUMNS SUMMARY:
               quantity    unit_price  total_amount        rating  return_days
count  14900.000000  15036.000000  15046.000000  15050.000000  1800.000000
mean       3.864966     18.046410     52.006810      3.913579    15.513333
std        2.211944     26.551215     89.005691      1.330433     8.523312
min       -5.000000      0.000000      0.000000     -1.000000     1.000000
25%        2.000000      3.709584     14.159753      3.000000     8.000000
50%        4.000000      9.087160     34.281709      4.000000    16.000000
75%        6.000000     21.143154     69.780811      5.000000    23.000000
max        7.000000    401.308912   4423.174301     10.000000    30.000000
```

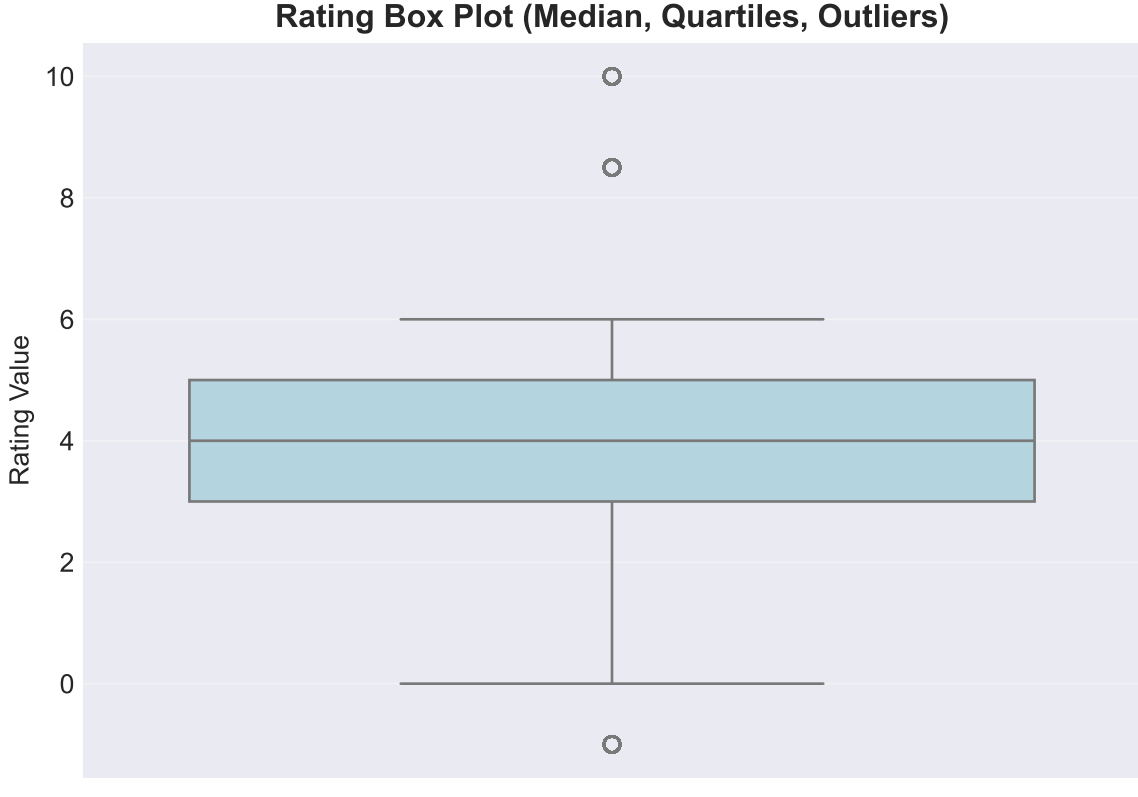
```
PLOT 1: HISTOGRAM - Rating Distribution

plt.figure(figsize=(10, 5))
plt.hist(copy_data['rating'], bins=20, color='steelblue', edgecolor='black', alpha=0.7)
plt.title('Distribution of Customer Ratings', fontsize=14, fontweight='bold')
plt.xlabel('Rating Value', fontsize=12)
plt.ylabel('Frequency (Count)', fontsize=12)
plt.grid(axis='y', alpha=0.3)
plt.show()

PURPOSE: Shows the distribution shape, central tendency, and spread of ratings.
```



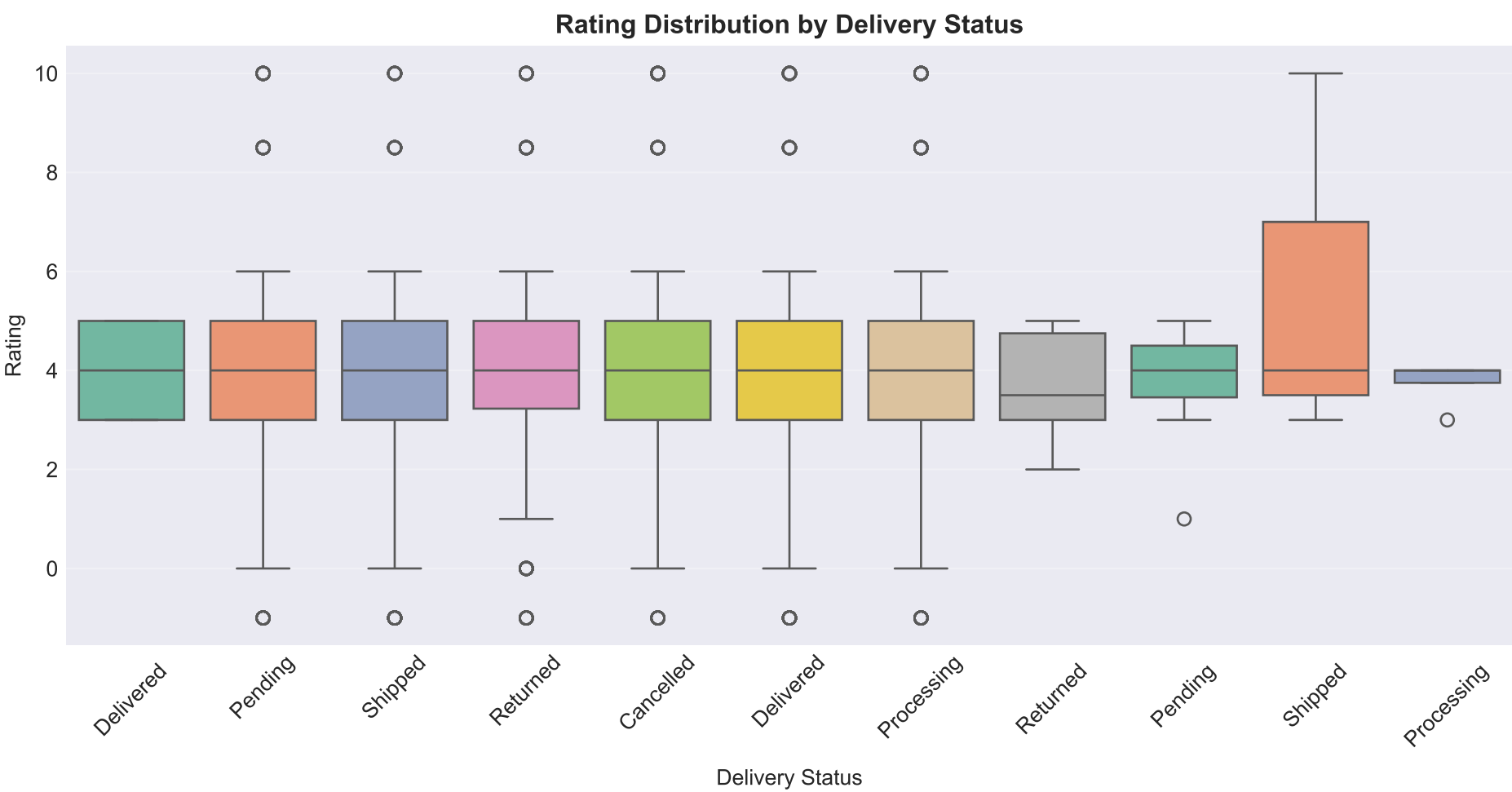
```
PLOT 2: HISTOGRAM WITH KDE - Smooth Distribution Curve

plt.figure(figsize=(10, 5))
sns.histplot(data=copy_data, x='rating', kde=True, bins=20, color='steelblue')
plt.title('Rating Distribution with KDE', fontsize=14, fontweight='bold')
plt.xlabel('Rating', fontsize=12)
plt.ylabel('Density', fontsize=12)
plt.show()

PURPOSE: KDE (Kernel Density Estimate) overlays a smooth curve showing probability distribution.
```



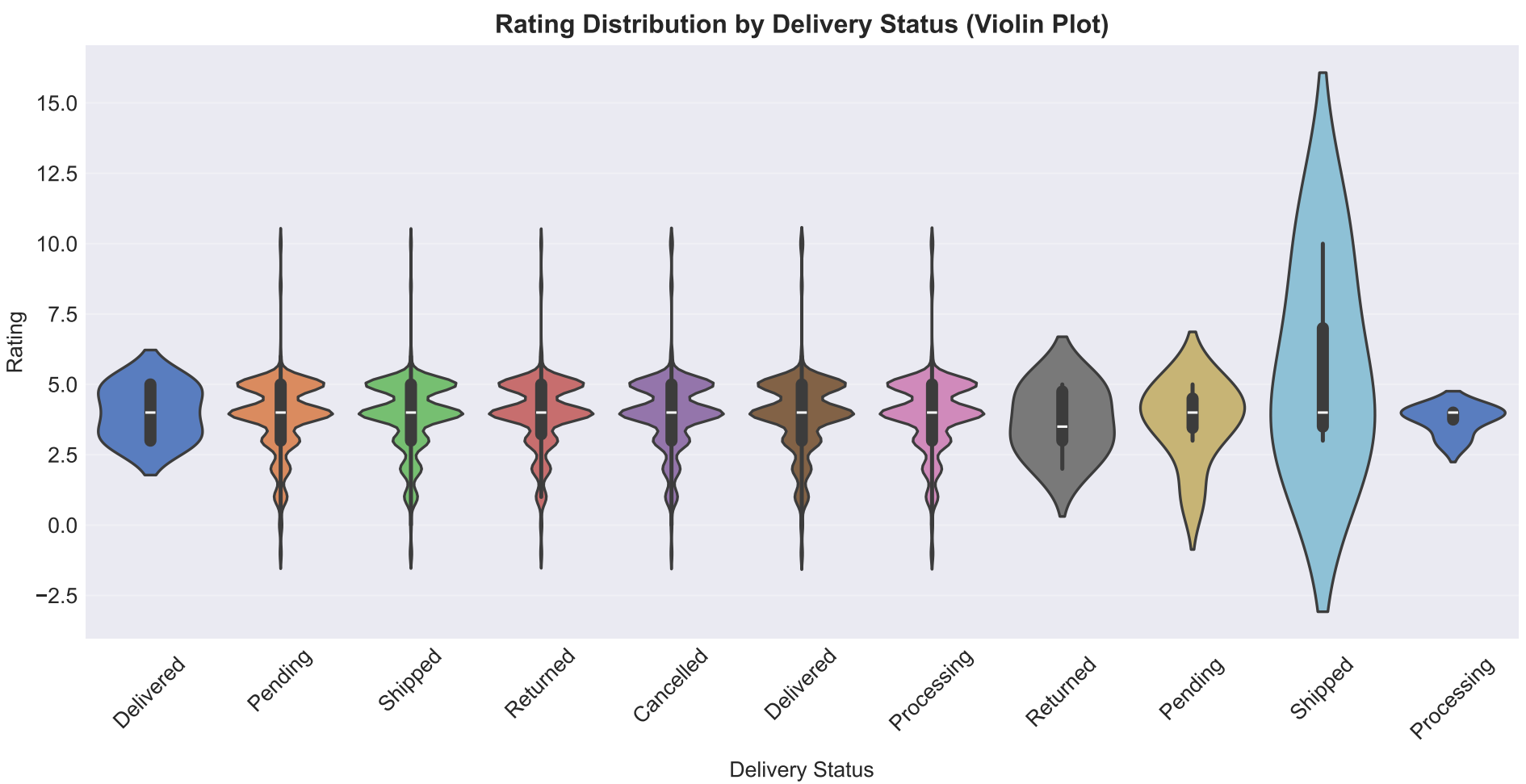
```
PLOT 3: BOX PLOT - Show Quartiles, Median & Outliers

plt.figure(figsize=(8, 5))
sns.boxplot(data=copy_data, y='rating', color='lightblue')
plt.title('Rating Box Plot', fontsize=14, fontweight='bold')
plt.ylabel('Rating Value', fontsize=12)
plt.grid(axis='y', alpha=0.3)
plt.show()

PURPOSE: Shows median (middle line), quartiles (box), and outliers (dots) for quick distribution understanding.
```



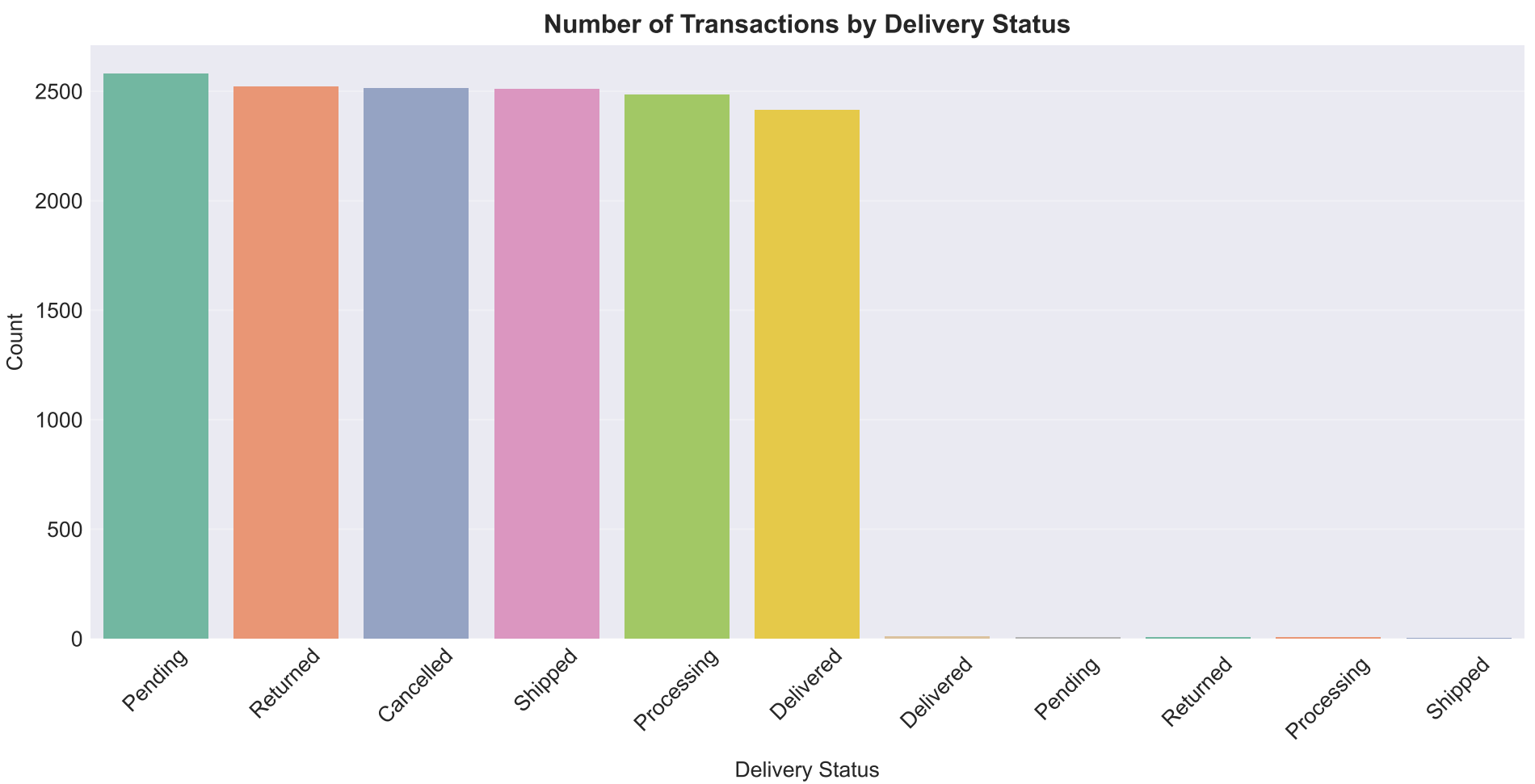
```
PLOT 4: GROUPED BOX PLOT - Compare Across Categories

plt.figure(figsize=(12, 6))
sns.boxplot(data=copy_data, x='delivery_status', y='rating', palette='Set2')
plt.title('Rating Distribution by Delivery Status', fontsize=14, fontweight='bold')
plt.xlabel('Delivery Status', fontsize=12)
plt.ylabel('Rating', fontsize=12)
plt.xticks(rotation=45, ha='right')
plt.grid(axis='y', alpha=0.3)
plt.show()

PURPOSE: Compare how ratings differ across delivery status categories.
```



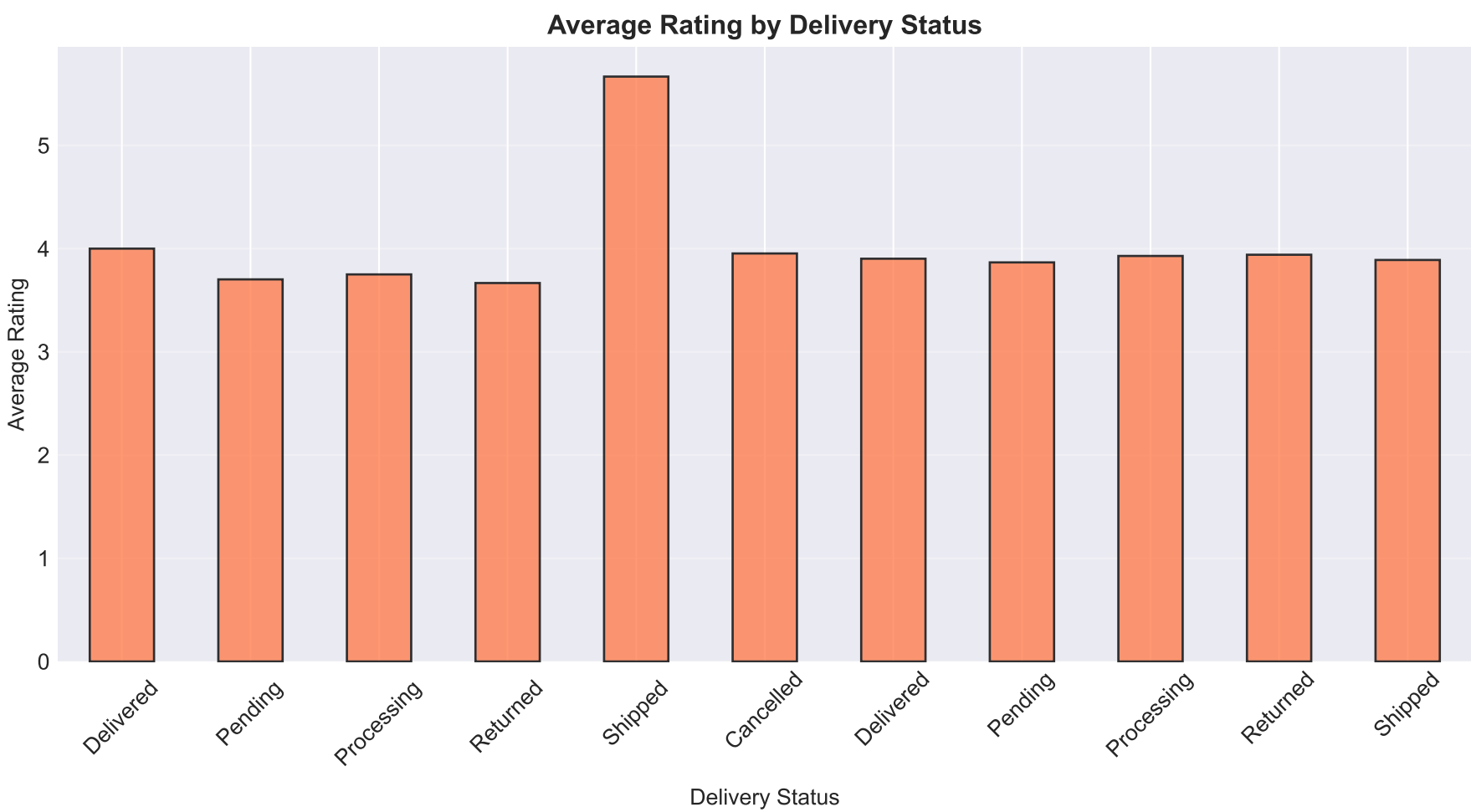
```
PLOT 5: VIOLIN PLOT - Show Full Distribution Density

plt.figure(figsize=(12, 6))
sns.violinplot(data=copy_data, x='delivery_status', y='rating', palette='muted')
plt.title('Rating Distribution by Delivery Status (Violin Plot)', fontsize=14, fontweight='bold')
plt.xlabel('Delivery Status', fontsize=12)
plt.ylabel('Rating', fontsize=12)
plt.xticks(rotation=45, ha='right')
plt.grid(axis='y', alpha=0.3)
plt.show()

PURPOSE: Shows full PDF (Probability Density Function) - wider sections = more customers at that rating.
```



```
PLOT 6: COUNT PLOT - Frequency of Each Category

plt.figure(figsize=(12, 5))
sns.countplot(data=copy_data, x='delivery_status', palette='Set2',
              order=copy_data['delivery_status'].value_counts().index)
plt.title('Number of Transactions by Delivery Status', fontsize=14, fontweight='bold')
plt.xlabel('Delivery Status', fontsize=12)
plt.ylabel('Count', fontsize=12)
plt.xticks(rotation=45, ha='right')
plt.grid(axis='y', alpha=0.3)
plt.show()

PURPOSE: Show how many transactions fall into each category.
```



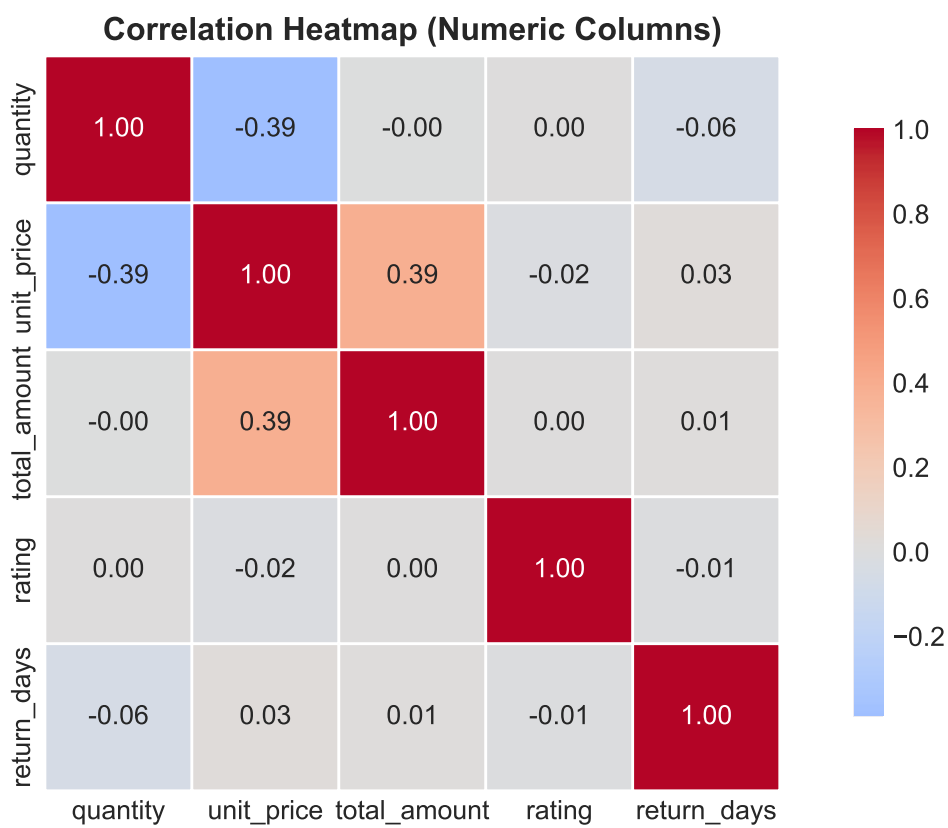
```
PLOT 7: BAR CHART - Average Values by Category

plt.figure(figsize=(10, 5))
avg_rating_by_status = copy_data.groupby('delivery_status')['rating'].mean()
avg_rating_by_status.plot(kind='bar', color='coral', edgecolor='black', alpha=0.8)
plt.title('Average Rating by Delivery Status', fontsize=14, fontweight='bold')
plt.xlabel('Delivery Status', fontsize=12)
plt.ylabel('Average Rating', fontsize=12)
plt.xticks(rotation=45, ha='right')
plt.grid(axis='y', alpha=0.3)
plt.show()

PURPOSE: Compare aggregated metrics (mean, sum, count) across categories.
```



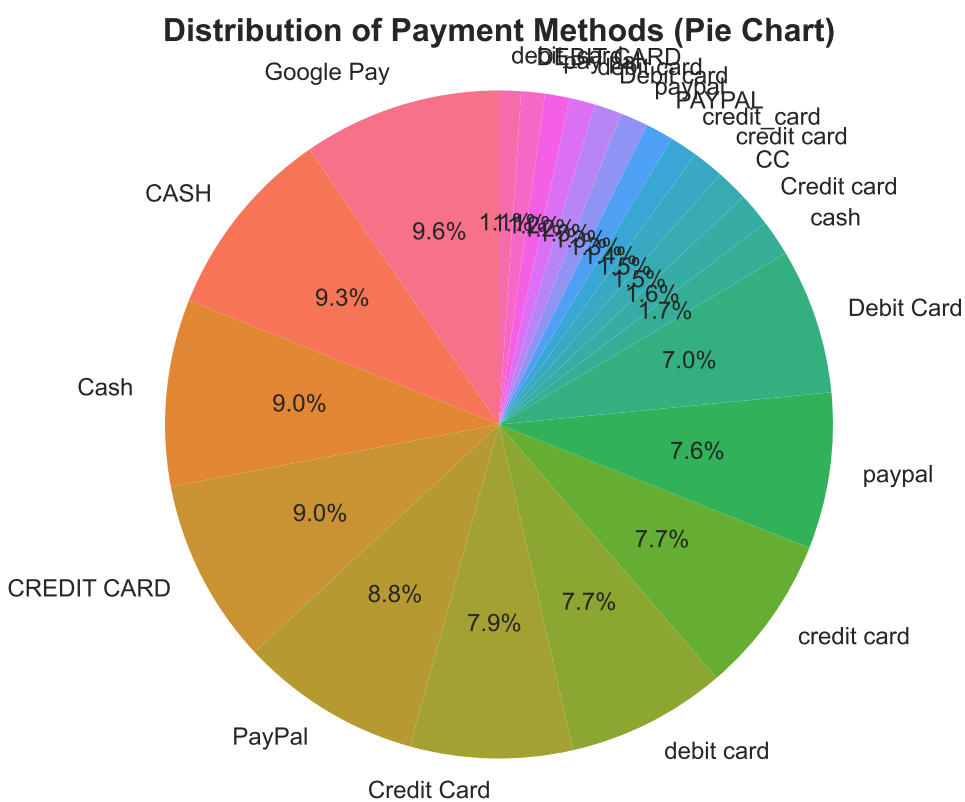
```
PLOT 8: SCATTER PLOT - Relationship Between Two Numeric Variables

plt.figure(figsize=(10, 6))
plt.scatter(copy_data['quantity'], copy_data['rating'], alpha=0.5, s=50, color='darkblue')
plt.title('Quantity vs Rating', fontsize=14, fontweight='bold')
plt.xlabel('Quantity Ordered', fontsize=12)
plt.ylabel('Customer Rating', fontsize=12)
plt.grid(alpha=0.3)
plt.show()

PURPOSE: Identify correlation - do variables trend together or inversely?
```

```
PLOT 9: SCATTER WITH HUE - Add Third Dimension with Color

plt.figure(figsize=(12, 6))
sns.scatterplot(data=copy_data, x='quantity', y='rating', hue='delivery_status', s=100, alpha=0.6)
plt.title('Quantity vs Rating (Colored by Delivery Status)', fontsize=14, fontweight='bold')
plt.xlabel('Quantity', fontsize=12)
plt.ylabel('Rating', fontsize=12)
plt.legend(title='Delivery Status', loc='best')
plt.grid(alpha=0.3)
plt.show()

PURPOSE: Visualize 3 dimensions: X-axis, Y-axis, and color (categorical variable).
```

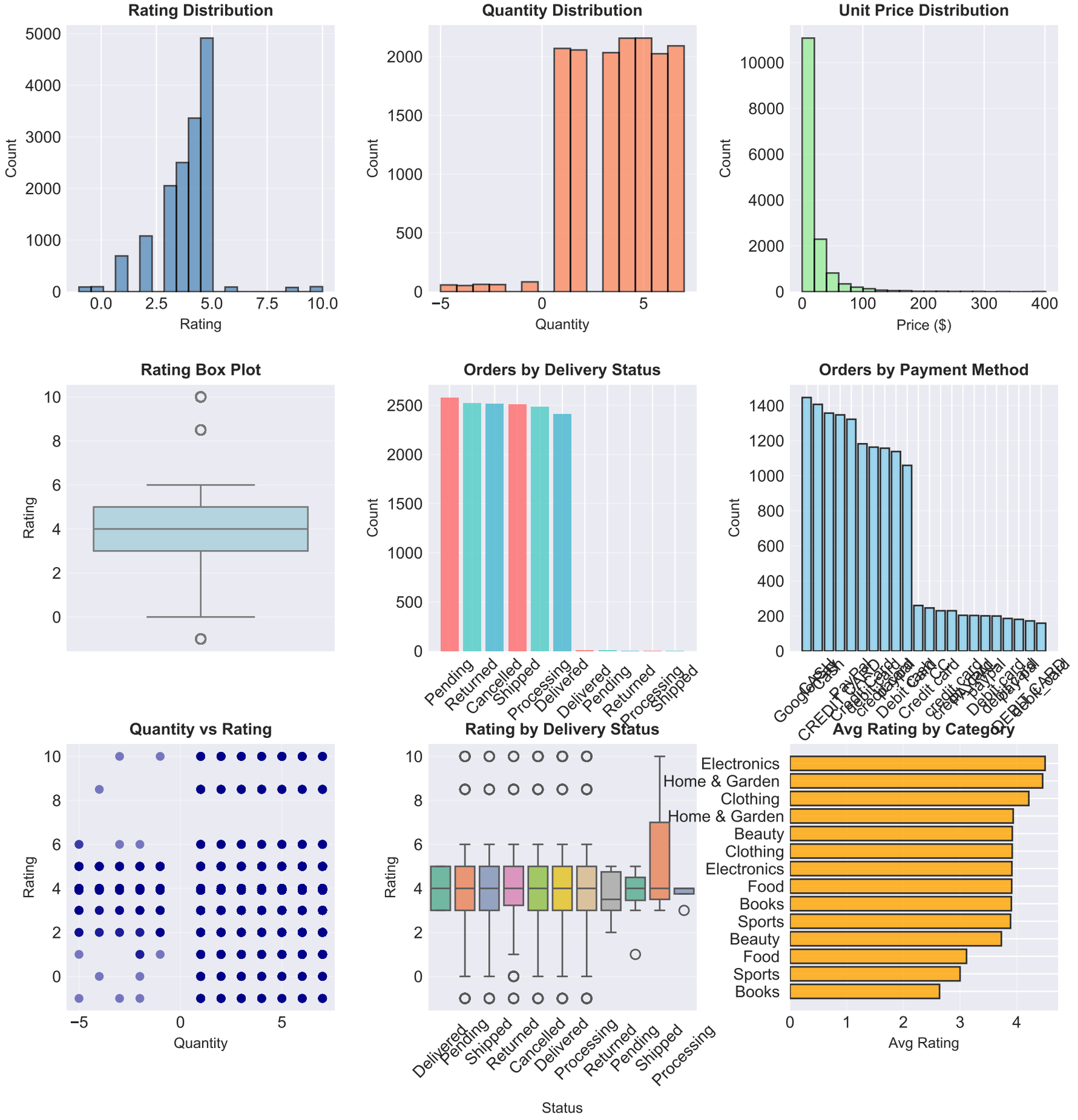
```
PLOT 10: CORRELATION HEATMAP - See All Numeric Relationships

plt.figure(figsize=(10, 8))
numeric_cols = copy_data.select_dtypes(include=['float64', 'int64']).columns
correlation_matrix = copy_data[numeric_cols].corr()
sns.heatmap(correlation_matrix, annot=True, fmt='.2f', cmap='coolwarm',
            center=0, square=True, linewidths=1, cbar_kws={"shrink": 0.8})
plt.title('Correlation Heatmap', fontsize=14, fontweight='bold')
plt.show()

PURPOSE: Red = positive correlation | Blue = negative correlation | White = no correlation
```



```
PLOT 11: PIE CHART - Show Proportions/Percentages

plt.figure(figsize=(10, 6))
payment_counts = copy_data['payment_method'].value_counts()
colors = sns.color_palette("husl", len(payment_counts))
plt.pie(payment_counts, labels=payment_counts.index, autopct='%1.1f%%',
        colors=colors, startangle=90, textprops={'fontsize': 11})
plt.title('Distribution of Payment Methods', fontsize=14, fontweight='bold')
plt.axis('equal')
plt.show()

PURPOSE: Show composition - which category makes up what % of the whole.
```



PLOT 12: E-Commerce Analysis Dashboard (9 Views)



KEY INSIGHTS & RECOMMENDATIONS

```
DATA OVERVIEW:
• Total Transactions: 15,050
• Average Rating: 3.91/5.0
• Average Order Quantity: 3.86 units
• Average Order Value: $52.01

DISTRIBUTION INSIGHTS:
• Most Common Rating: 5.0
• Rating Range: -1.0 to 10.0
• Most Common Delivery Status: Pending
• Most Popular Payment Method: Google Pay

MISSING DATA HANDLED:
• Customer IDs: 752 filled with 'Unknown'
• Ratings: 0 NaN (all filled with mean = 3.91)
• Return Days: 13,250 missing (88% of data)

RECOMMENDATIONS FOR VISUALIZATION:
1. Use Histograms → Understand distribution shapes
2. Use Box Plots → Identify outliers and compare groups
3. Use Scatter Plots → Find correlations
4. Use Count/Bar Plots → Compare categorical frequencies
5. Use Heatmaps → Visualize all relationships at once
6. Use Pie Charts → Show composition/percentages
7. Use Dashboards → Combine multiple views for comprehensive analysis

BUSINESS INSIGHTS:
• Delivery Status Distribution shows patterns in order fulfillment
• Rating distribution reveals customer satisfaction levels
• Payment method preferences indicate transaction patterns
• Scatter plots help identify operational efficiencies

NEXT STEPS:
1. Filter data by specific time periods or categories
2. Segment customers by rating to identify VIP vs at-risk customers
3. Analyze payment method preferences by delivery status
4. Investigate low ratings to identify quality issues
5. Use time-series analysis if date data is available
```